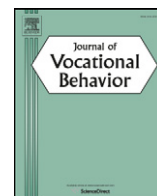


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A transactional approach to person-environment fit: Reciprocal relations between personality development and career role growth across young to middle adulthood

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ABSTRACT

In order to enhance our understanding of person–environment transactions, the present longitudinal cohort study examined the dynamic interactions between career role development and personality development over a time interval of 15 years. A sample of college alumni ($N = 260$) provided self-reports on Big five traits three months prior to graduation and 15 years later when their career had unfolded. In addition, detailed descriptions of their career role trajectories over this time interval were obtained. Results first indicated significant positive associations between personality trait levels and initial career role engagement: Extraversion predicted Presenter, Director, Inspirator and Guide roles; Conscientiousness predicted Expert role; Agreeableness predicted Guide role; and Openness to experience predicted Presenter role. Further, initial trait levels were found to predict subsequent changes in career role engagement, and the strength of these associations varied according to career stage (first versus second half of the time interval). Finally, change in career roles over time was associated with change in at least one personality trait, except Openness to experience. The study provides insights into the largely unstudied question of how career identities grow and how this relates to personality development during young adulthood.

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Introduction

The construct of person–environment (PE) fit is considered a cornerstone in the field of vocational psychology (Savickas, 2000), and is rooted trait and factor theory. Accordingly, choosing an occupation involves trying to match an individual to a job so that their needs will be met and their job performance will be satisfactory (Sharf, 1992). Whereas the application of PE fit in trait and factor theory relied heavily on the measurement of individual traits and (static) matching processes with environmental requirements, more contemporary perspectives on PE fit have introduced the concept of *dynamic reciprocity* (Rounds & Tracey, 1990). This indicates an ongoing concept of adjustment as environments are influenced by individuals and individuals are influenced by environments. “The P × E fit perspective explicitly assumes that people and environments change continually in ongoing adjustment” (Chartrand, 1991, p. 521). This significant concept addresses some of the important criticisms that have been leveled at trait and factor theory (Patton & McMahon, 1999). Most importantly, the advantage of this dynamic perspective on PE fit over static trait and factor theory is that it offers a way for understanding the *career development process*.

Looking at PE fit research, however, very little effort has been done over the past decades in order to gain insight into dynamic reciprocity. Instead, studies addressing PE fit typically adopt a static approach in which fit or congruence is related to career outcomes assessed at the same or a later point in time (Ishitani, 2010). Although there are some studies that have adopted a more dynamic approach by incorporating *change* (e.g., Donohue, 2006; Oleski & Subich, 1996), these are severely restricted in that they only consider environmental change (i.e. job mobility) as a way to enhance fit.

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The present study aims to contribute to research on PE fit by focusing on the reciprocal relations between P and E characteristics. Drawing on current perspectives from personality psychology regarding trait development and using a recently developed career roles model to conceptualize career development, the present study presents a comprehensive test of *person–environment transactions* in a longitudinal sample of working adults spanning the first 15 years of the professional career.

Neo-socioanalytic perspectives on trait development

Five-Factor Model (FFM) personality traits (McCrae & Costa, 1987) are used in the present study to conceptualize the P component of PE transactions. These traits have been successfully related to different aspects of vocational behavior, including work environment preferences (Larson, Rottinghaus, & Borgen, 2002), nature of employment (e.g., De Fruyt & Mervielde, 1999), and career path characteristics (e.g., Wille, De Fruyt, & Feys, 2010). Further, from a developmental perspective, compelling evidence has accumulated over the past decade indicating that change in personality traits is possible at all ages across the life course, with the preponderance of change occurring in young adulthood (Roberts, Robins, Caspi, & Trzesniewski, 2003). Normative developmental changes, such as the tendency of people to become more agreeable, conscientious, emotionally stable and socially dominant have been observed across birth cohorts, national cultures, and using both longitudinal and cross-sectional research designs (Roberts, Walton, & Viechtbauer, 2006).

Interestingly, research has also indicated meaningful interindividual differences in personality development (Roberts, Caspi, & Moffitt, 2003), indicating that not all people follow these normative tendencies to the same extent, or even in the same direction. This has led to the hypothesis that life experiences, including those at work, play a significant role in personality trait change. Within the neo-socioanalytic model of personality (Roberts & Wood, 2006), personality traits and occupational experiences are in constant transaction across the life span. The present study corroborates this reciprocal hypothesis by exploring (a) the role that personality plays in shaping individuals' careers and (b) the impact of individuals' careers on personality development.

Career roles as units of analysis

While former theoretical approaches of careers (e.g., Super; 1980) assumed largely fixed and normative career stages, more recent work describes a career as a dynamic and interactive process of continuous learning (Hall & Mirvis, 1995). Hence, the notion of development has become more central in career theory, and this is elaborated in developmental contextual approaches such as the life-span developmental contextual theory of career development formulated by Vondracek and colleagues (Vondracek & Kawasaki, 1995; Vondracek, Lerner, & Schulenberg, 1986). Central in this theory is the notion of dynamic interaction, in relation to both development in general and vocational development in particular, that captures the way in which contexts and persons interact, influence, and transform one another.

From this new tradition, Hoekstra (2011) introduced the Career Roles Model (CRM) as an interesting new theoretical framework to understand career development in terms of emerging career role identity. Accordingly, the essential issue for understanding the nature and development of individual careers concerns the underlying roles. Broadly defined, roles and role labels serve a communicative purpose to capture a complex mixture of tasks, goals, values, norms, and expectations concerning a position in some social context (Ashforth, 2001). In the case of work roles, this context is work and Hoekstra (2011) uses the concept *career roles* to describe “those enduring aspects of work roles that the person identifies with and is identified with” (p. 5). According to the CRM, career roles are thus to be considered as the building blocks of individual careers, and a taxonomy is provided that distinguishes between six roles (*Maker, Expert, Presenter, Guide, Director, and Inspirator*) that are valid and potentially attainable in most jobs with some employee autonomy (Hoekstra, 2011).

Career role development

Beyond the identification of career roles as the building blocks of individual careers, the CRM also outlines a detailed perspective on career development. Accordingly, career development is the gradual and interactive acquisition of a repertoire of career roles over time. It is a gradual process, as career development spans the entire work life; it is interactive, as career development is the result of two simultaneous forces. On the one hand, individuals select certain roles based on their personal preferences and competencies (i.e., role taking); on the other, people are driven toward certain roles as a result of environmental demands and expectations (i.e., role pressure). Over the years, roles and commitments are selectively strengthened, while at the same time people keep adapting to external pressures and changes by innovating and negotiating their roles. Career development is, thus, to be understood as processes of *identity construction* in the work life (Hoekstra, 2011). The result of this process is an emerging career identity, which is to be considered as a continually updated internalization of career roles and associated commitments and aspirations.

Based on these general outlines, we believe there are two important reasons why the CRM is particularly appealing to study PE transactions. First, the model explicitly acknowledges the role the individual plays in shaping his or her career. This is true for the content of career development as well as for shape. Specifically, the individual is expected to pursue a career that fits his or her personal constellation. Furthermore, while some individuals prefer their role repertory to remain stable and unchanged during long periods of time, others are more engaged in exploring new roads and allow their career identity to continually grow and change. A second reason to argue for the CRM in the context of PE transactions refers to its underlying processes. The CRM explicitly refers to the process of identity development to describe career development. This is appealing, as it has been argued that role identities may offer

a way to understand how life experiences, including those at work, and general personality traits interact over time (Wood & Roberts, 2006).

PE transactions: Attraction–Selection–Transformation–Manipulation–Attrition

Drawing on Schneider's Attraction–Selection–Attrition framework (ASA; Schneider, 1987), Roberts (2006) introduced an elaborated Attraction–Selection–Transformation–Manipulation–Attrition (ASTMA) model that allows for a thorough understanding of person–environment transactions in the work context. The processes described by the ASTMA model form the basis of the hypotheses examined in the present study.

Attraction and selection

The first two mechanisms described by the ASTMA model deal with the employment process itself. On the one hand, *Attraction* refers to the process by which people are attracted to work environments that fit with their personality. *Selection* refers to the fact that people are also selected into and oriented towards specific work environments that are supposed to be compatible with their personal characteristics.

Role engagement, according to the CRM, is supposed to result from two simultaneous forces, i.e. role taking and role pressure, that closely align with both mechanisms described above. Role taking describes the process through which individual employees are attracted to specific career roles, and Hoekstra (2011) hereby draws on socioanalytic personality theory (Hogan & Roberts, 2000) to classify the principal motives that drive people to certain work environments. The second process, role pressure, is generally described as a counterforce that represents the demands and expectations from the environment (Hoekstra, 2011). Employees are selectively mobilized to ensure organizational stability, which entails selecting and orienting employees' with the primary purpose of continued membership (p. 163). Employers have well-defined expectations about an individual's role in the organization and about the competencies that are required to succeed. Evidence for the role that personal characteristics play in these processes of role taking (attraction) and role pressure (selection) comes from research demonstrating the validity of personal dispositions (e.g. personality traits and vocational interests) to predict status and nature of employment (e.g., De Fruyt & Mervielde, 1999).

Although no prior research has examined the associations between personality traits and actual career role engagement, a number of expectations can be formulated based on the content of each of the six roles. In the present study, these mechanisms of attraction and selection are examined by associating participants' Big Five personality traits, measured at the very beginning of the professional career, to their initial levels of role engagement early in the career.

Engagement in both the roles of Maker and Expert is assumed to be driven by goals regarding personal mastery and success, and a strong emphasis in these roles is put on independent individual production (Hoekstra, 2011). In terms of personality characteristics, these role descriptions most closely align with the Conscientiousness domain, which has also been labeled as the 'Will to achieve' (Digman & Inouye, 1986). Therefore, Conscientiousness scores at the beginning of the career are expected to be positively associated with early career engagement in the Maker and Expert roles (*Hypothesis 1a*).

The Presenter and Guide roles both emphasize the social interactions with others (Hoekstra, 2011). As Agreeableness and Extraversion are considered to be particularly relevant in the context of interpersonal behavior (Wiggins & Broughton, 1985), these traits can be expected to relate to engagement in these roles. In the Presenter role, social interactions are typically used with the primary purpose of convincing and influencing others. Clearly, these interactions require an energetic and dominant position, and therefore it is expected that Extraversion scores at the beginning of the career are positively related to early career engagement in the Presenter role (*Hypothesis 1b*). On the other hand, the Guide role focuses on connecting with and committing to others, interactions which require stronger empathic competencies. Therefore, Agreeableness scores at the beginning of the career are expected to be positively associated with early career engagement in the Guide role (*Hypothesis 1c*).

Finally, the roles of Director and Inspirator share their focus on the collective development of groups and organizations (Hoekstra, 2011). Whether it is with the purpose of realizing long term goals (Director role) or to initiate strategic change (Inspirator role), it can be assumed that both roles have a high decision latitude in common. As research has demonstrated that the degree of decision making opportunities at work correlates positively with Extraversion (e.g., Grant & Langan-Fox, 2006), it is expected that Extraversion scores at the beginning of the career are positively associated with early career engagement in Director and Inspirator roles (*Hypothesis 1d*).

Attrition and manipulation

The ASTMA model further describes two ways by which individuals can actively shape their work environment. First, it is argued that people can decide to leave their work environment, a mechanism referred to as *Attrition* (Schneider, Smith, Taylor, & Fleenor, 1998). In addition, *Manipulation* refers to the idea that individuals consciously as well as unconsciously may attempt to modify their organizational experiences in order to maximize fit (Roberts, 2006). This is in line with the idea of job crafting which suggests that individuals can affect their day-to-day work experience by altering the tasks they do, organizing their work differently, or by changing the nature of the relationships they maintain with others (Wrzesniewski & Dutton, 2001).

According to the CRM, career development is day-to-day micro development, expressed in the gradual strengthening, weakening, and change of certain roles (Hoekstra, 2011). In the present study, individuals' attempts to shape their work environment are studied by examining the degree of stability and change in career role engagement, over and within jobs, across the first fifteen years of the professional career. Based on prior research documenting the long-term stability of work environments (e.g., Sutin & Costa, 2010) we first expect to find moderate levels of rank order stability in role engagement across the entire time interval (*Hypothesis 2*).

In addition, more specific expectations can be formulated about the effects of personality scores at the beginning of the career and subsequent *change* in career role engagement. A recurrent assumption in person-environment theories is that individuals shape their work environment, whether it is by changing employers or by modifying job content, in the pursuit of a better fit. Sutin and Costa (2010), for example, found that individuals high on Emotional Stability had jobs characterized by higher levels of decision latitude, and that emotionally stable individuals actively shaped their jobs over the next 10 years to include even more decision making latitude. The findings suggested that as individuals progress through their careers in midlife, they mold their everyday experiences on the job to fit their personality. Consistent with these findings, we expect the concurrent associations between traits and role engagement at the beginning of the career to be reflected in the pattern of trait effects on change in career role engagement (*Hypothesis 3*). For example, if we succeed in finding a positive association between Agreeableness and initial levels of Guide role engagement at the beginning of the career, then we also expect individuals high on Agreeableness to increase in Guide role engagement throughout the next 15 years of their career.

Transformation

Finally, the ASTMA model also allows for people to change themselves in response to occupational experiences, a process referred to as *Transformation* (Roberts, 2006). This is undoubtedly the most innovative aspect of the ASTMA model, and draws on the growing body of literature demonstrating some effects of occupational experiences on personality trait development (for an overview of these studies, see Roberts, 2006). While several studies have demonstrated the effect of work role *quality* (i.e., satisfaction) on personality trait change (e.g., Scollon & Diener, 2006), research on the effects of work role *content* is still scarce and the established empirical evidence remains modest (Sutin & Costa, 2010).

In the present study, we address the issue of transformation by examining the associations between individual differences in personality trait change and career role engagement. Prior theorizing about these effects (Roberts & Caspi, 2003) has proposed the corresponsive principle as a testable hypothesis to examine transformation mechanisms (*Hypothesis 4*). Accordingly, the predominant effects of work experience on changes in personality traits will be on those traits that predict the specific work experience. Consistent with earlier research on reciprocal effects between personality traits and work experiences (e.g., Roberts, Caspi, & Moffitt, 2003), the corresponsive hypothesis is tested in the present study by examining whether the personality traits that predict career role engagement are the same traits that show change in these same career roles. It is expected that patterns of predictive relationships between traits and career roles, reflected in the static, concurrent correlations, will be mirrored in the relationships between changes in traits and changes in career roles. For example, if early career engagement in the Guide role is, in part, predicted by higher levels of Agreeableness, then Agreeableness will be the disposition most likely to change in response to increasing Guide role engagement over time.

Method

Participants

To test these hypotheses, data are used from a longitudinal research project on personality development, labor market entrance, and career unfolding in a large sample of Flemish college alumni with measurement occasions in 1994, 1995, 2009, and 2010 (De Fruyt & Mervielde, 1999; Wille et al., 2010). Specifically, data are used from a subsample of 260 participants (137 males and 123 females) who provided personality data in 1994 (T1) and 2009 (T2), and who also provided a complete overview of their career unfolding across the 15-year time interval between both assessment points. Participants' mean age was 22.8 years ($SD = 1.07$) at T1 and 37.9 years ($SD = 1.07$) at T2. Although all participants in this study were highly educated, there was substantial heterogeneity in terms of their vocational preferences. Specifically, the alumni included in this study represented various college and university faculties, including Philosophy, History and Languages ($n = 44$), Law ($n = 27$), Industrial engineering ($n = 58$), Sciences ($n = 19$), Applied Sciences ($n = 33$), Economics ($n = 22$), Psychology and Educational Sciences ($n = 30$), Applied Biological Sciences ($n = 6$), and Political and Social Sciences ($n = 21$). We examined the participants' job titles to get a picture of the type of work they typically fulfilled at T2. There was a large variety of jobs, with occupations ranging from fashion designer to academic professor. The mean number of working hours per week was 41.08 ($SD = 10.77$).

The issue of dropout is inherent to longitudinal studies, especially when time intervals are large. Originally, 934 final year college students participated in the current alumni research project. To test for selectivity in dropout at T2, continuers ($n = 260$) were compared to dropouts ($n = 674$) with regard to their personality scores at T1. The results however showed no significant mean differences between both groups in terms of Big Five scores, $F(5, 928) = 1.45, p > .05$.

Instruments

Personality (T1 and T2)

At both T1 and T2, FFM personality traits were assessed using the Dutch adaptation of the NEO-PI-R (Costa & McCrae, 1992; Hoekstra, Ormel, & De Fruyt, 1996). The Dutch Revised NEO Inventory is a faithful translation of the NEO-PI-R, with a factor structure and psychometric properties closely resembling the normative US Inventory (De Fruyt & Mervielde, 1997). For the present study, internal consistencies of FFM domain scores (T1/T2) were .91/.92 for Neuroticism, .90/.90 for Extraversion, .86/.88 for Openness, .91/.90 for Agreeableness, and .90/.91 for Conscientiousness.

Career role development (T2)

Using an interactive web application, semi-standardized reports of participants' career role trajectories (from September 1994 until April 2009) were obtained in two stages. First, they were asked to break this career stage down into successive time intervals according to job and/or organizational changes (for a thorough overview of this methodology, see Wille et al., 2010). In brief, each time unit had to be specified with a starting and ending date and covered at least three months. Respondents were asked to classify each interval as a particular work (e.g., new job with a new employer) or non-work (e.g., career interruption due to sickness) stage, and were invited to report complementary descriptive information (e.g., job titles). In a second phase of the online questionnaire, participants were given an overview of their entire career trajectory as entered in the first stage, and were subsequently asked to score each of the work stages in terms of the importance of each of the six career roles. For this purpose, they were presented a shortened version of the Dutch Career Roles Questionnaire (CRQ; Hoekstra, 2011) to evaluate each work stage. A shortened version was used in order to reduce the testing load for the respondents, given the presumed relatively large number of career changes across these 15 years for some individuals. The questionnaire consisted of 18 items; three items for each of the six career roles, with each item describing an illustrative example of functioning in one career role (see Appendix A). Respondents were asked to indicate on a 7-point scale how well the statement described the role they typically had in their work during that specific work stage. For the non-work stages, scores on the CRQ were automatically fixed at zero.

In this manner, respondents provided a highly personalized overview of their career role engagement with the total number of work-related career phases as well as the length of these phases varying across and within individuals. This dataset was subsequently reorganized toward a time structured dataset with career role scores for each individual on a pre-defined number of measurement occasions. First, the 15-year time interval (from September 1994 until April 2009) was subdivided into 29 semesters, and each respondent was given career role scores for each semester based on their personal trajectories. When necessary, career role scores of 2 individual career stages were averaged to retain a single semester score. Career interruptions were treated as missing values for the semesters they spanned, using the Full Information Maximum Likelihood approach (FIML; Schafer & Graham, 2002) available in Lisrel 8.7 (Jöreskog & Sörbom, 2004). Finally, the number of measurement occasions was further reduced by averaging career role scores across five (only for the first measurement occasion) or four semesters, resulting in seven career stages and accompanying measurement points for each respondent (T1 = September 1994–February 1997; T2 = March 1997–February 1999; T3 = March 1999–February 2001; T4 = March 2001–February 2003; T5 = March 2003–February 2005; T6 = March 2005–February 2007; T7 = March 2007–February 2009). Compared to other solutions (six or eight measurement points), seven measurement points proved to be the best balance between data reduction and information loss. Exploratory (EFA) and confirmatory (CFA) factor analyses were conducted in order to establish the construct validity of the shortened career roles instrument, using CRQ data that described participants' initial work-related experiences (i.e. first jobs). This enabled us to test the construct validity in the largest sample available. An EFA (principal component analysis and varimax rotation) first indicated that a six-factor solution accounted for 78.6 percent of the variance. Factor loadings indicated that all items loaded primarily and substantially (>.60) on their intended factor. Although some items, particularly those from the inspirator-domain, also demonstrated cross-loadings, these did not exceed .40. Next, a CFA on the shortened CRQ, specifying a measurement model with all items loading on their intended factor, indicated good to acceptable fit indices (normed $\chi^2 = 2.98$, CFI = .93, RMSEA = .07, SRMR = .06). Together, these results provide strong evidence for the construct validity of the shortened CRQ. Internal consistencies of the six career roles, averaged across the seven measurement occasions, were .75 (Director), .82 (Inspirator), .82 (Presenter), .88 (Guide), .81 (Maker), and .93 (Expert).

Statistical analyses

Preliminary analyses

First, a series of preliminary analyses were performed to explore general indicators of personality trait and career role development. Mean-level stability of observed scores was examined using repeated measures ANOVAs and associated standardized mean differences (*d* effect sizes). In addition, test-retest correlations were computed to assess the level of relative stability in traits and roles across time.

Latent variable analyses

Latent Change Models (LCMs; McArdle & Nesselroade, 1994) were used to model absolute change in Big Five traits over the two measurements of personality across the 15-year interval. The major advantage of LCMs is that initial level and change are directly estimated as latent variables for each personality factor. This makes it possible to estimate the variance of the latent change factor as a parameter. Rejecting the null hypothesis of zero variance indicates that there are reliable interindividual differences in change. Another attractive feature of these LCMs is that they can also be used in structural models in which change in one variable is related to change in another (Hertzog, Dixon, Hultsch, & MacDonald, 2003). In order to avoid overall model complexity, we trimmed down the number of observed indicators by combining the items of two personality facets into one parcel set. For example, for Neuroticism, the items of N1 Anxiety and N2 Angry Hostility were merged into one parcel, like the items of N3 Depression and N4 Self-Consciousness, and the items of N5 Impulsiveness and N6 Vulnerability. In this manner, each of the Big Five traits was modeled longitudinally with three observed indicators defining the associated latent variables.

The availability of more than three measurements of career role engagement allowed us to apply Latent Growth Curve Models (LGMs; Bollen & Curran, 2006) to examine career development over time. LGMs can address two issues of particular interest for the present study: description of trajectories in career role engagement over time and identification of interindividual variation in the rate of change in career roles.

Finally, having identified the appropriate ways to model personality and career role development separately, we subsequently combined both techniques in a series of 30 (five traits $\times$ six roles) multivariate structural equation models to examine the transactions between both developmental patterns. Correlations of particular interest include the correlations between initial trait levels and subsequent career role slopes which indicate individuals' attempts to shape their work environment. Furthermore, patterns of correlated change (i.e., correlations between trait change factors and career role slopes) illustrate how individuals change in relation to changing career roles. All latent variable analyses were conducted using Lisrel 8.7 (Jöreskog & Sörbom, 2004).

Results

Trait development

General patterns of personality trait change over the 15 year time interval were first examined at the level of the observed variables. With regard to rank-order stability, the results indicated moderate to high levels of continuity with test-retest correlations equal to .56 for Agreeableness and Conscientiousness, .63 for Openness, .65 for Neuroticism, and .67 for Extraversion. All test-retest correlations were significant at $p < .001$. Regarding mean-level change, the results demonstrated significant decreases in Neuroticism ($d = -.47, p < .001$) and Openness ($d = -.37, p < .001$) between 1994 and 2009. On the other hand, participants significantly increased in Conscientiousness ($d = .46, p < .001$) and Agreeableness ($d = .12, p < .05$). No significant mean-level change was found for Extraversion ($d = -.06, p > .05$).

In a second step, individual level personality trait change was examined using LCMs (see Table 1). These models fitted the data well ($RMSEA < .08$; Browne & Cudeck, 1993), except for Conscientiousness ($RMSEA = .10$). Nonetheless, comparative fit indices indicated excellent model fits (NFI , $NNFI$, and $CFI > .95$) for all traits, including Conscientiousness.

Of particular interest in these LCMs are the variances associated with the latent change factors, addressing the issue of interindividual differences in personality change. As shown in Table 1, the results showed significant levels of interindividual differences in trait change for each of the Big Five traits. The negative correlations between initial level and change further indicated that individuals who were initially higher on the personality dimensions exhibited greater decreases (or smaller increases for Neuroticism, Openness, and Extraversion) across the follow-up period.

Career role development

Parallel to the Big Five trait analyses, developmental patterns in career role engagement were initially examined at the level of the observed variables. First, and consistent with our expectations (Hypothesis 2), career roles showed moderate to high levels of rank order stability over the entire 15-year interval. Test-retest correlations were .38 (Expert), .45 (Director), .47 (Inspirator), .48 (Guide), .51 (Maker), and .62 (Presenter). All test-retest correlations were significant at $p < .001$.

A series of repeated measures ANOVAs were conducted to assess the level of mean-level change in career role engagement between two successive career stages and across the entire time interval (see Table 2). Standardized mean differences first indicated significant increases in each of the six career roles over the 15-year interval ($d_{1,7}$ varied between .54 and 1.32, $p < .001$). Further, inspection of the effect sizes associated with career role change between two successive time intervals ($d_{t,t+1}$) suggested nonlinear growth in career roles over time, with stronger increases from T1 to T4 compared to the marginal increments from T4 to T7.

Table 1

Summary of the latent change models for the big five traits.

Model	Model fit				Latent variable	Change-level
	<i>NFI</i>	<i>NNFI</i>	<i>CFI</i>	<i>RMSEA</i>	Variance	Correlation
Neuroticism	.99	.98	.99	.06	0.85 [†]	-.34 ^{**}
Level					0.49 [†]	
Change						
Extraversion	1.00	1.00	1.00	.00	0.81 [†]	-.42 ^{**}
Level					0.42 ^{**}	
Change						
Openness	1.00	1.00	1.00	.00	0.75 [†]	-.26 [*]
Level					0.50 ^{**}	
Change						
Agreeableness	.99	1.00	1.00	.03	0.69 [†]	-.49 ^{**}
Level					0.46 [†]	
Change						
Conscientiousness	.98	.96	.98	.10	0.45 [†]	-.43 ^{**}
Level					0.31 [†]	
Change						

Note. Latent variable analyses are based on the entire sample ($N = 260$).

^{**} $p < .01$.

^{*} $p < .05$.

[†] $p < .001$.

Table 2

Mean-level and rank order stability of career roles across all seven assessment points.

	T1		T2				T3				T4				T5				T6				T7				
	Sep '94		Mar '97				Mar '99				Mar '01				Mar '03				Mar '05				Mar '07				
	–		–				–				–				–				–				–				
	Feb '97		Feb '99				Feb '01				Feb '03				Feb '05				Feb '07				Feb '09				
	M	SD	M	SD	d _{1,2}	r _{1,2}	M	SD	d _{2,3}	r _{1,3}	M	SD	d _{3,4}	r _{1,4}	M	SD	d _{4,5}	r _{1,5}	M	SD	d _{5,6}	r _{1,6}	M	SD	d _{6,7}	d _{1,7}	r _{1,7}
Expert	4.20	1.81	4.71	1.71	.29 [†]	.80 [†]	5.04	1.58	.20 [†]	.60 [†]	5.38	1.45	.23 [†]	.52 [†]	5.45	1.46	.05	.44 [†]	5.42	1.54	–.02	.42 [†]	5.52	1.51	.07 [*]	.79 [†]	.38 [†]
Maker	5.16	1.39	5.41	1.39	.18 [†]	.80 [†]	5.66	1.27	.19 [†]	.68 [†]	5.78	1.17	.10 [*]	.52 [†]	5.79	1.18	.01	.50 [†]	5.78	1.22	–.01	.52 [†]	5.85	1.16	.06 [*]	.54 [†]	.51 [†]
Presenter	3.43	1.63	3.83	1.73	.24 [†]	.85 [†]	4.11	1.76	.16 [†]	.77 [†]	4.44	1.72	.19 [†]	.66 [†]	4.59	1.69	.09 ^{**}	.65 [†]	4.66	1.71	.04	.62 [†]	4.81	1.70	.09 [†]	.83 [†]	.62 [†]
Guide	3.81	1.68	4.15	1.67	.20 [†]	.80 [†]	4.53	1.65	.23 [†]	.66 [†]	4.90	1.60	.23 [†]	.55 [†]	5.02	1.55	.08 [*]	.54 [†]	5.07	1.56	.03	.50 [†]	5.21	1.51	.09 [†]	.88 [†]	.48 [†]
Director	2.99	1.50	3.64	1.66	.41 [†]	.75 [†]	4.08	1.65	.27 [†]	.60 [†]	4.51	1.67	.26 [†]	.45 [†]	4.74	1.62	.14 [†]	.44 [†]	4.88	1.67	.08 [*]	.41 [†]	5.06	1.64	.11 ^{**}	1.32 [†]	.45 [†]
Inspirator	3.87	1.50	4.29	1.52	.28 [†]	.81 [†]	4.57	1.49	.18 [†]	.68 [†]	4.92	1.41	.24 [†]	.56 [†]	5.03	1.38	.08 ^{**}	.52 [†]	5.14	1.35	.09 [*]	.49 [†]	5.24	1.34	.07 [*]	.97 [†]	.47 [†]

Note. Career role scores are computed on a scale from 1 to 7. Test–retest correlations r indicate the relative or rank order stability of career role engagement between two assessment points. Standardized difference scores d represent the effect sizes for mean-level change. Due to missing values, N ranges between 243 and 248 for test–retest correlations and between 243 and 258 for mean-level changes.

[†] $p < .001$.

^{**} $p < .01$.

^{*} $p < .05$.

Next, individual level career role change was further analyzed using a latent variable approach. Considering the nonlinear trend in career role growth, piecewise growth models (e.g., Bollen & Curran, 2006) were established to model change in each of the career roles over the 15-year interval (see Fig. 1). Piecewise trajectory modeling is meant “to approximate the nonlinear function through the use of two or more linear piecewise splines” (Bollen & Curran, 2006, p. 103). For this purpose, time was coded in a way that expressed piecewise linear change, placing the origin of the time scale (i.e., 0) at the fourth measurement occasion, namely the point at which the steeper increases in each of the career roles changed to less pronounced growth. The slope loadings associated with T1 through T3 (i.e., -3 , -2 , and -1) indicate the assumed linear growth during the first career stage. Similarly, the slope loadings after T4 (i.e., 1 , 2 , and 3) define linear growth in career roles during the second stage. These piecewise LGMs fitted the data very well, with RMSEA fit indices varying between .00 and .03.

A summary of these piecewise LGMs is given in Table 3. Inspection of the slope means indeed indicated stronger increases in role engagement during the first compared to the second career half. In addition, mean increases in the Expert and Maker role failed to reach significance during the second career stage. Significant slope variances furthermore indicated interindividual differences in role change during both career stages. The positive correlations between initial slopes and intercepts illustrated that stronger increases in role engagement during the first career stage were associated with higher levels in these roles at T4. On the contrary, the negative correlations between career role levels at T4 and subsequent role changes suggested smaller increases in roles for those scoring higher on these roles at T4. Finally, the negative correlation between both slopes indicated that, except for the Expert role, larger increases in role engagement during the first career stage were associated with smaller increases in these roles during the second stage.

Relations between traits and initial career role engagement

Associations between personality and career role engagement were first addressed by examining the concurrent correlations between T1 Big Five traits and retrospective evaluations of career role engagement during the first career stage. The pattern of correlations largely supported our hypotheses on attraction and selection effects. For Conscientiousness, one significant association was found: as expected (Hypothesis 1a), Conscientiousness scores at the beginning of the career were positively related to initial engagement in the Expert role ($r = .13$, $p < .05$). The expected association between Conscientiousness and engagement in the Maker role could not be confirmed ($r = .08$, $p > .05$). In line with our expectations, individuals initially scoring higher on Extraversion reported stronger early career engagement in the Presenter (Hypothesis 1b; $r = .19$, $p < .01$), Director (Hypothesis 1d; $r = .16$, $p < .05$) and Inspirator (Hypothesis 1d; $r = .19$, $p < .01$) roles. No significant correlations were found between initial Extraversion and Expert ($r = .03$, $p > .05$) and Maker ($r = .06$, $p > .05$) roles.

Finally, one significant association was found for Agreeableness. Consistent with our expectations (Hypothesis 1c), individuals initially scoring higher on Agreeableness also reported stronger engagement in the Guide role during the first career stage ($r = .17$, $p < .01$).

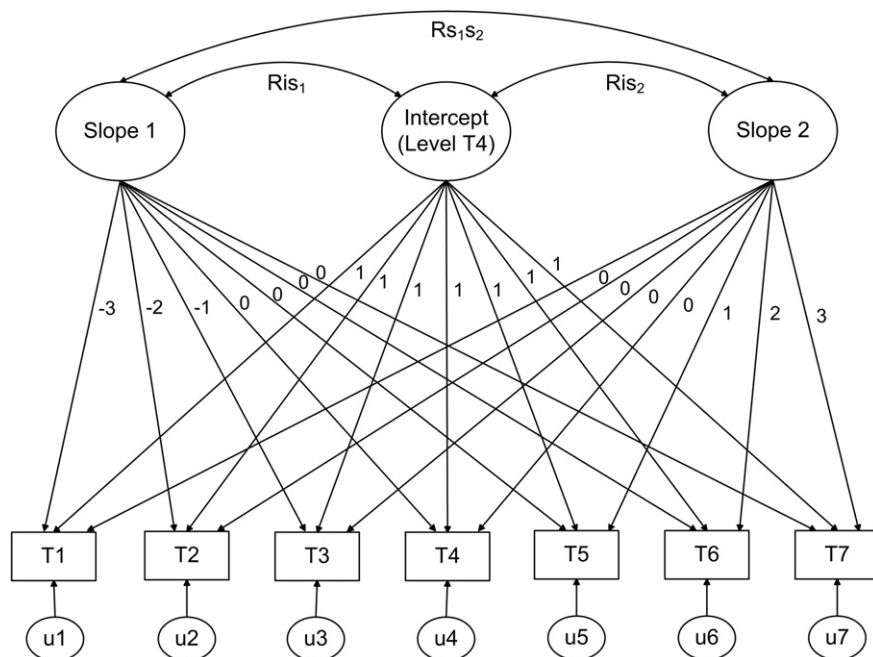


Fig. 1. Piecewise latent growth model of career role change across seven measurement occasions (T1 through T7). The model incorporates a latent intercept and two slopes. Note that the intercept in this model represents the fourth measurement occasion; Slope 1 represents change in career role engagement prior to T4; Slope 2 represents change after T4. u_1 through u_7 indicate random error associated with the observed data. The paths from the latent slopes to the observed scores imply two linear growth functions, one prior to T4 (loadings -3 , -2 , and -1) and one after T4 (loadings 1 , 2 , and 3). The relationship between the intercept and the slopes is represented by the correlations between the intercept and the slopes (Ris_1 and Ris_2). Rs_{1s_2} represents the relationship between both slopes.

Table 3

Summary of the piecewise latent growth curves for the career roles.

Model	Model fit RMSEA	Latent variable mean	Latent variable variance	Slope1-intercept <i>r</i>	Intercept-slope2 <i>r</i>	Slope1-slope2 <i>r</i>
Expert	.02					
Slope1		0.39 [†]	0.16 [†]	.38 [†]	–.25 ^{**}	–.05
Intercept		5.36	2.08			
Slope2		0.04	0.12 [†]			
Maker	.02					
Slope1		0.20 [†]	0.18 [†]	.38 [†]	–.38 [†]	–.34 [†]
Intercept		5.75	1.49			
Slope2		0.03	0.08 [†]			
Presenter	.03					
Slope1		0.33 [†]	0.23 [†]	.43 [†]	–.28 ^{**}	–.25 ^{**}
Intercept		4.44	2.29			
Slope2		0.12 [†]	0.10 [†]			
Guide	.00					
Slope1		0.36 [†]	0.24 [†]	.41 [†]	–.35 [†]	–.22 [*]
Intercept		4.89	2.38			
Slope2		0.11 [†]	0.13 [†]			
Director	.00					
Slope1		0.50 [†]	0.31 [†]	.59 [†]	–.42 [†]	–.41 [†]
Intercept		4.52	2.74			
Slope2		0.18 [†]	0.12 [†]			
Inspirator	.03					
Slope1		0.33 [†]	0.20 [†]	.42 [†]	–.48 [†]	–.24 [*]
Intercept		4.91	2.00			
Slope2		0.11 [†]	0.06 [†]			

Note. Latent variable analyses are based on the entire sample ($N = 260$).[†] $p < .001$.^{**} $p < .01$.^{*} $p < .05$.

Two significant correlations were found that were not a priori expected. Higher Openness scores at the beginning of the career were associated with stronger engagement in the Presenter role ($r = .17, p < .01$) and scores on Extraversion correlated with initial Guide role engagement ($r = .25, p < .01$). Finally, no significant associations were found between initial levels of Neuroticism and early career role engagement.

Trait effects on career role development

Longitudinal associations between personality traits and career roles were examined using latent variable models that combined trait LCMs and role LGMs (see Fig. 2). These models produced acceptable model fits, with RMSEA indices varying between .03 and .08.

Trait effects on career role development were examined by testing the structural associations between trait levels (representing personality scores at the beginning of the career) on the one hand and career role Slope 1 (representing role change between T1 and T3), role intercept (representing the level of career role engagement at T4), and Slope 2 (representing role change between T5 and T7)

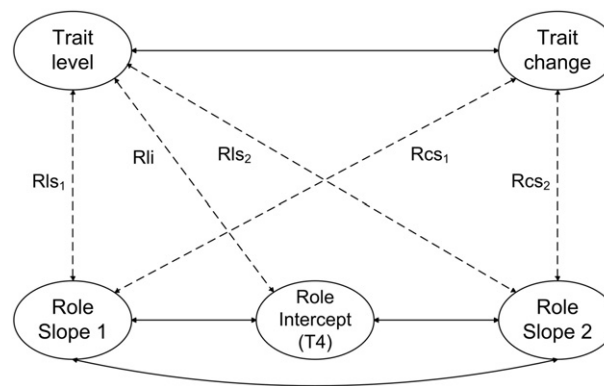


Fig. 2. Combined structural equation model to test transactions between personality and career role development over time. Dotted lines indicate the structural relations of interest: Rls_1 and Rls_2 represent the correlation between initial Trait level and change in career role engagement during the first (from T1 to T3) and second (from T5 to T7) career stage respectively. Rli represents the correlation between initial Trait level and level of career role engagement at T4. Rcs_1 and Rcs_2 represent the correlation between Trait change and change in career role engagement during the first and the second career stage respectively.

on the other. Building on prior research addressing personality effects on the work environment, we expected the concurrent associations between traits and roles at the beginning of the career to be reflected in trait effects on career role change (*Hypothesis 3*). Based on the significant concurrent associations between traits and roles at the beginning of the career, trait effects on role development were thus expected for Extraversion, Openness, Agreeableness, and Conscientiousness. The results of these analyses are shown in columns two to six of Table 4 (“Level”).

For Extraversion, it was expected that higher trait levels at the beginning of the career were associated with stronger increases in the Presenter, Guide, Director, and Inspirator roles. The results, however, showed no significant associations between level of Extraversion and change in Presenter, Director, and Inspirator roles during both career stages. For the Guide role, we found a significant negative association between Extraversion level and role change during the first career stage ($r = -.15, p < .05$). Extraverts showed smaller increases in this role, whereas more introverted people had stronger increases. Nonetheless, extraverts still scored significantly higher on the Guide role at T4 ($r = .15, p < .05$). Similarly, the results showed that level of Extraversion remained a significant predictor of Presenter ($r = .13, p < .05$), Director ($r = .14, p < .05$) and Inspirator ($r = .21, p < .01$) role scores at T4.

Based on the concurrent association at the career start, we further expected the level of Openness to be associated with change in Presenter role engagement. The results showed that individuals initially scoring higher on Openness did not show stronger increases in the Presenter role during the first ($r = -.01, p > .05$) or the second ($r = .05, p > .05$) career stage, but nevertheless maintained to score higher on this role at T4 compared to individuals initially lower on Openness ($r = .14, p < .05$). Unexpectedly, individuals scoring higher on Openness showed stronger increases in the Expert role ($r = .13, p < .05$), although this effect only reached significance for role change during the second career stage. Similarly, we found a positive effect of initial Openness scores on increases in Maker role during this second career stage ($r = .17, p < .05$).

For Agreeableness, we hypothesized that individuals scoring high on this trait at the beginning of the career would show stronger increases in the Guide role over time. The results, however, did not show significant trait effects on Guide role slopes during the first ($r = -.01, p > .05$) or the second ($r = -.07, p > .05$) career stage, although higher scores on Agreeableness at the beginning of the career were still associated with higher scores on the Guide role at T4 ($r = .19, p < .01$). Unexpectedly, we also found that individuals initially scoring higher on Agreeableness showed stronger increases in the Director role during the first career stage ($r = .15, p < .05$).

Individuals scoring higher on Conscientiousness at the beginning of the career did not show stronger increases in the Expert role during the first career stage ($r = .00, p > .05$), but nevertheless continued to engage more strongly in this role at T4 compared to individuals low on Conscientiousness ($r = .13, p < .05$). Contrary to what could be expected, we found a significant negative association between level of Conscientiousness and Expert role slope during the second career stage ($r = -.15, p < .05$). This indicated that individuals initially lower on Conscientiousness showed greater increases in the Expert role during the second career stage.

Table 4

Correlations representing the structural relations in the combined SEM models.

Roles	Traits									
	Level					Change				
	N	E	O	A	C	N	E	O	A	C
Expert										
Slope1	.04	.09	.15	.06	.00	.06	-.15	-.04	-.08	-.08
Intercept	.02	-.03	.00	-.02	.13*					
Slope2	-.03	.03	.13*	-.02	-.15*	.02	.06	.03	.07	.14*
Maker										
Slope1	.22**	-.07	.06	-.06	-.04	-.07	.12	-.07	.02	-.01
Intercept	.10	-.04	.05	-.03	.04					
Slope2	-.04	-.04	.17*	-.04	.00	.00	.06	-.03	.14*	-.05
Presenter										
Slope1	.04	-.01	-.01	.00	-.01	-.04	-.02	-.06	-.05	.10
Intercept	-.03	.13*	.14*	-.04	-.12					
Slope2	-.06	.00	.05	-.06	-.03	-.16*	.16*	.08	.11	.08
Guide										
Slope1	-.09	-.15*	-.03	-.01	-.08	.13	.12	.09	-.04	.14*
Intercept	.00	.15*	.07	.19**	-.01					
Slope2	-.02	.05	-.01	-.07	-.05	-.08	.00	-.07	.06	.10
Director										
Slope1	-.11	.06	.00	.15*	-.04	.10	-.01	-.03	-.24**	.13
Intercept	-.11	.14*	-.04	.10	-.01					
Slope2	.01	.05	.13	-.13	-.15*	-.25**	.07	-.05	.17*	.25**
Inspirator										
Slope1	.00	.08	-.02	.06	-.07	-.02	-.05	-.03	-.15*	.15*
Intercept	.02	.21**	.11	.04	-.03					
Slope2	-.07	.12	.11	.03	-.17*	-.18*	-.05	.02	.08	.26**

Note. Latent variable analyses are based on the entire sample ($N = 260$).

* $p < .05$.

** $p < .01$.

compared to high scorers. Furthermore, the results unexpectedly showed that individuals initially lower on Conscientiousness showed stronger increases in the Director ($r = -.15, p < .05$) and Inspirator ($r = -.17, p < .05$) roles during the second career stage.

Finally, no trait effects on role change were expected for Neuroticism given the lack of concurrent associations at the beginning of the career. Nonetheless, we found one significant effect on career role change: higher scores on Neuroticism at the beginning of the career were associated with stronger increases in the Maker role during the first career stage ($r = .22, p < .01$). At T4, the high scorers on Neuroticism had evolved toward a Maker role level that low scorers had reached earlier, as reflected in the insignificant association between Neuroticism and T4 Maker role scores ($r = .10, p > .05$).

Relations between change in traits and change in roles

The combined latent variable models were further used to examine associations between change in traits and change in career roles over the 15-year interval. Correlated change was examined by testing the structural associations between personality change factors (representing trait change over the entire 15-year interval) on the one hand and career role slopes on the other. In accordance with the corresponsive principle, we hypothesized that these patterns of correlated change would also reflect the pattern of concurrent associations between traits and career roles at the beginning of the career (*Hypothesis 4*).

We first specifically expected change in Extraversion to be related to change in Presenter, Guide, Director, and Inspirator roles. However, as can be seen in Table 4 (columns “Change”), this was only supported for the Presenter role, with higher increases in Extraversion being associated with higher increases in this role during the second career stage ($r = .16, p < .05$).

Contrary to our expectations, change in Openness was not significantly related to change in the Presenter role during the first ($r = -.06, p > .05$) or the second ($r = .08, p > .05$) career stage. In fact, our results showed that change in Openness was not associated with change in any of the six career roles.

The expected association between change in Agreeableness and change in Guide role also failed to reach significance ($r = -.04, p > .05$ and $r = .06, p > .05$ for the first and second career stage respectively). Change in Agreeableness nonetheless correlated significantly with change in three other career roles. Particularly, increases in Agreeableness were positively associated with increases in the Maker ($r = .14, p < .05$) and Director ($r = .17, p < .05$) roles during the second career stage. In addition, negative associations were found between change in Agreeableness and change in Director ($r = -.24, p < .01$) and change in Inspirator ($r = -.15, p < .05$) roles during the first career stage.

Consistent with the concurrent relations at the beginning of the career, we found a significant positive association between increases in Conscientiousness and growth in the Expert role ($r = .14, p < .05$), although this effect was restricted to role change during the second career stage. Change in Conscientiousness was furthermore positively associated with change in the Guide role (first career stage; $r = .14, p < .05$), change in the Director role (second career stage; $r = .25, p < .01$), and change in the Inspirator role (first and second career stage; $r = .15, p < .05$ and $r = .26, p < .01$ respectively).

Finally, as scores on Neuroticism were not predictive of engagement in any of the career roles, we did not expect to find evidence for correlated change for this trait. Nonetheless, our results demonstrated that change in Neuroticism was significantly and negatively related to change in Presenter ($r = -.16, p < .05$), Director ($r = -.25, p < .01$) and Inspirator ($r = -.18, p < .05$) roles, although these effects were restricted to the second career stage.

Discussion

Personality trait development

The period of young adulthood has been described as an influential stage of life with regard to personality development. Despite the moderate to high levels of rank-order stability in traits that have also been reported previously (e.g., Roberts & DelVecchio, 2000), the results of the present study further substantiate the notion of personality development by documenting on mean-level changes that are in line with the normative developmental patterns (i.e., increases in Conscientiousness and Agreeableness; decreases in Neuroticism and Openness) reported elsewhere (Roberts, Walton, & Viechtbauer, 2006). Further, the latent variable analyses additionally indicated that each of the Big Five traits also showed significant individual differences in change across the 15-year interval. This strengthens the idea that environmental influences, like vocational experiences, can influence individual's trait change during young adulthood.

Career role development

Not only is young adulthood a crucial period in individuals' personal lives, it also imposes a number of defining challenges in the professional sphere. One of the attractive features of this study includes the possibility to follow participants across what are probably the most essential years in their vocational development starting from career entrance. To our knowledge, no prior research has examined career trajectories in working adults over such a long time interval and with such great detail. We found the CRM particularly appealing for this purpose, as it was explicitly designed for vocational development purposes (Hoekstra, 2011).

Our results indicated substantial mean-level increases in role engagement for each of the six career roles. Furthermore, stronger increases in career roles were found during the first half of the interval compared to the increases during the second half. This nonlinearity in role growth suggests the existence of two different career stages during this period of vocational development. Using Super's (1980) career stage model to describe this type of normative career progression, the first career stage of steeper career role growth indicates the

phase of *establishment*, with individuals growing to a specific career identity. The more stable role trajectories over the next seven years conversely suggest a phase of *maintenance*, in which the career identity is further consolidated.

Latent variable analyses also yielded significant variances in the career role slopes, suggesting that not all participants followed these mean-level trajectories to the same extent or even in the same direction. Growth curves further demonstrated that individuals who increased more in role engagement during the first career stage, showed smaller increases during the second stage and vice versa. This indicated a developmental trend toward homogeneity in the sample, with individual differences between participants in terms of career role engagement gradually declining over time. One explanation for this phenomenon may lie in the fact that the participants included in this study were all highly educated, which could explain their shared advancement toward occupational positions similar in terms of underlying roles.

Person–environment transactions

A large body of research in the vocational development literature addresses the issue of person–environment fit, assuming that (a) people choose occupational activities that fit with their personal characteristics, and (b) people simultaneously are selected into work environments based on their personal characteristics. Building on these ideas, we hypothesized that traits –through processes of either attraction or selection– would predict initial career role engagement measured early in the career. The present study was the first to show significant relations between Big Five traits and career role engagement. We specifically found traits, assessed at the very beginning of the career, to be related to retrospective evaluations of initial career role engagement. Moreover, these associations were largely consistent with our hypotheses that assumed a striving for person–environment fit, although the obtained associations were typically modest in magnitude. Individuals scoring high on Conscientiousness reported stronger engagement in the Expert role which is driven by personal motives regarding personal mastery, success, and well-being. Extraverts scored higher on the Presenter and Guide roles, both centered around social interaction, and were also more likely to engage in Director and Inspirator roles, which are typically associated with more decision making opportunities. Stronger altruistic competencies were predictive of stronger Guide role engagement which requires well-developed empathy skills. Finally, an unexpected association was observed between scores on Openness and engagement levels in the Presenter role. According to Hoekstra (2011), the Presenter role focuses on persuasion by means of interpersonal effectiveness with form, style, and impression management. The positive association with Openness possibly illustrates the more creative component of this role.

Interestingly, Neuroticism did not relate significantly to initial engagement in any of the career roles. Although career roles and interests are not the same thing, the observed null findings involving Neuroticism are consistent with research that has found nonsignificant relations between Neuroticism and vocational interests (Larson et al., 2002).

Besides concurrent associations between traits and roles at the beginning of the career, we also expected traits to influence the subsequent pattern of career role development in an attempt to maximize person–environment fit. To examine this, personality trait scores were used to predict career role growth scores during the next 15 years, over and within jobs. We specifically expected individuals to further develop those career roles that they initially preferred based on their baseline personality characteristics.

Although several significant associations were found, evidence for trait effects on role change was weak and the associations that were identified did not correspond with this developmental hypothesis. In general, we found that the concurrent associations between traits and roles at the beginning of the career were replicated for (static) career role scores in the middle of the 15 year interval. For example, individuals scoring higher on Agreeableness not only reported higher Guide role scores at the career start but also seven years later. However, traits did not predict the role change trajectories as we expected. In the case of Agreeableness, no positive effects were found between initial trait scores and Guide role change during the first or the second career stage.

Taken together, these findings challenge the idea that individuals selectively *increase* in those aspects of the work role that are compatible with their personal characteristics. Two comments can be made here. First, the results may suggest that the idea of “changing the work environment in the pursuit of a satisfying person–environment fit” operates differently when this work environment is defined in terms of the underlying roles. More specifically, it seems that individuals are more motivated to *maintain* a satisfying role position over time rather than striving for *disproportionate increases* in desired roles. Second, it is important to bear in mind that individual differences in career role trajectories are also to a certain extent environmentally determined. Remember that Hoekstra (2011) defines role development as an interactive process of both role taking and role pressure, the latter referring to the impact of external expectations (from supervisors, coworkers,...) and economic necessities. In this regard, it is not uncommon that individuals, through the complex process of career progression, are pushed toward certain occupational positions and associated roles that do not entirely align with their own psychological make-up.

The final mechanism of person–environment transaction that we discuss is transformation, referring to the idea that experiences in professional settings can promote change in personality traits (Roberts, 2006). Consistent with prior research on personality development and work experiences, we expected the traits that “selected” people into specific career roles, whatever the process, to be the same traits that are subject to change in response to those same career roles (Roberts, Caspi, & Moffitt, 2003). Although the results of this study provided further evidence for patterns of correlated change, this was hardly in accordance with the corresponding principle.

Change in career role engagement was associated with change in four of the five personality traits; only for Openness, no significant associations were found between trait change and change in career role engagement. The results importantly indicate that *increases in career role engagement generally promote normative personality trait development*. For example, normative decreases in Neuroticism are more pronounced in individuals showing stronger increases in Director, Presenter and Inspirator roles. Similarly, normative increases in Conscientiousness are more pronounced in individuals showing stronger increases in Director, Inspirator, Guide, and Expert roles. This

is consistent with other research demonstrating that investment in the work role contributes to normative personality development (e.g., Roberts, Caspi, & Moffitt, 2003).

However, the results also illustrate how investment in certain aspects of the work role can contribute to *non-normative* trait development (i.e., changes in personality traits that run counter to general trends). Specifically, we found that stronger increases in the Director and Inspirator role during the first career stage were associated with smaller increases or even decreases in Agreeableness. Apparently, these two career roles impose certain behaviors or tendencies to people that buffer or hinder the naturally expected growth in Agreeableness. Prior research has demonstrated that non-normative trait change in young adulthood is associated with de-investment in the work role, like engagement in counterproductive work behaviors (Roberts, Walton, Bogg, & Caspi, 2006). The results of the present study provide interesting new insights in this domain by showing that stronger work role involvement not necessarily contributes to normative personality development, but that the effect depends on the specific work role content.

Limitations and future research

A number of limitations to this study should be noted. First, our investigation of personality trait change was limited as only two measurement occasions were available. Although LCMs offer a powerful technique to study trait change in such a design (McArdle & Nesselrode, 1994), they are limited as they implicitly assume linear change between both measurement occasions. An additional personality assessment in the middle of the 15-year interval may have revealed whether stronger career role change during the first career stage was simultaneously associated with higher levels of trait change in that period.

This study was the first to relate trajectories of work role content to patterns of personality development over the same time interval. To this end, we asked participants to retrospectively report on the importance of each of the six career roles over the past 15 years. We are aware that retrospective evaluations may be partly biased or inaccurate. Nonetheless, the interactive web application that we used for this purpose maximally allowed participants to give reliable and valid descriptions of their career role development. Participants were first asked to reconstruct their own personal career trajectory, based on (objective) job and/or organizational changes. Next, they were asked to reflect on each of these stages by (a) giving a short description of them and (b) evaluating them in terms of the six career roles operationalized in terms of fundamental work characteristics that can easily be remembered. Participants thus reconstructed a personal time line defined by individually relevant marking points. This kind of methodology has been positively evaluated in life course research (Berney & Blane, 1997), particularly since recall has been shown to be substantially improved by introducing a personal temporal reference system for the subject to use (Means, Swan, Jobe, & Esposito, 1991). Of course, we promote future studies addressing dynamic reciprocity using more prospective longitudinal designs.

We chose Hoekstra's (2011) model of career roles to examine PE transactions given its association with identity development in work life, bridging both personal and vocational development. We encourage future researchers to continue exploring PE transactions hereby drawing on other established career models, such as Holland's (1985) RIASOC theory.

Conclusion

Drawing on the success of trait and factor theory, person-environment fit has been proposed as a refined career theory that also incorporated the idea of reciprocity between P and E characteristics. In the present study, we took this evolution a step further and explicitly examined the underlying dynamics of PE fit. Using recent developments from both personality and vocational psychology, we argued for a transactional perspective which proposes trait effects on career unfolding and vocational effects on trait development. We introduced the concept of *career identity*, defined in terms of the roles one identifies with and is identified with, as a mediating mechanism between both developmental processes. We believe that this transactional perspective on PE fit contributes to a more comprehensive understanding of the career development process. Further, it opens the door for future innovative research examining the longitudinal and dynamic interplay between personal and vocational characteristics.

Appendix A. Shortened version of the Career Roles Questionnaire

			Describes my role						
			1 Not at all 7 Very well						
			1	2	3	4	5	6	7
1.	Give shape to ideas and images in an appealing way.	(P)							
2.	Apply expert knowledge in new problems.	(E)							
3.	Pay attention to someone's personal story.	(G)							
4.	Invigorate and provoke others with challenging views.	(I)							
5.	Press on firmly to deliver an assignment.	(M)							
6.	Take the lead in confusing situations.	(D)							
7.	Have a plan approved by framing it in the proper way.	(P)							
8.	Shed a light on a subject by informed comment.	(E)							
9.	Achieve something with a person by empathic understanding.	(G)							
10.	Show my vision of things at work in an enthusiastic way.	(I)							

(continued on next page)

Appendix A (continued)

			Describes my role						
			1 Not at all 7 Very well						
			1	2	3	4	5	6	7
11.	Organize matters personally to attain good results.	(M)							
12.	Phrase the decision after a discussion.	(D)							
13.	Dress up a plan or product to make it attractive and convincing.	(P)							
14.	Analyze a problem that others find complicated.	(E)							
15.	Gain someone's confidence in a delicate matter.	(G)							
16.	Transmit thoughts about essential values in our work.	(I)							
17.	Spare no effort when the real work has to be done.	(M)							
18.	Manage a change process.	(D)							

(P) = Presenter, (E) = Expert, (G) = Guide, (I) = Inspirator, (M) = Maker, (D) = Director.

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